

Alessandro Petitti

ROBOTICS AND CONTROL ENGINEER

Rome, Italy / Zürich, Switzerland

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Education

ETH - Eidgenössische Technische Hochschule

Zürich, Switzerland

MSC IN ROBOTICS SYSTEMS AND CONTROLS

Sep 2024 - present

- GPA: 5.45/6
- Relevant coursework: Physically based Simulation in computer graphics, Machine Perception, Vision Algorithms for Mobile Robotics, Robot Dynamics, Nonlinear Dynamics and Control, Model Predictive Control, Rehabilitation Engineering.

Università Campus Bio-Medico di Roma

Rome, Italy

BSC. IN INDUSTRIAL ENGINEERING

Sep 2021 - July 2024

- GPA: 30.17/30, graduated with full honors
- Relevant coursework: Applied Mechanics, Rational Mechanics Laboratory, Mechanics of solids and structures, Electronics, Bioengineering laboratory, Control of Digital Systems, Operating Systems and Computer Networks

Experience

Robocup team member - ETH Zurich

IFA - Automatic Control Laboratory

TEAM MEMBER

Jan 2025 - Sep. 2025

- Part of the robo-soccer team, in the motion control and state estimation area, with the goal of making standardized robot play football autonomously.

EPFL - Ecole Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

RESEARCH INTERN

July - Sep 2024

- As a participant in the highly selective E3 program at EPFL, with a 2% acceptance rate, I was engaged in advanced research under the supervision of Dr. Josie Hughes. My work focused on soft robotics, particularly in soft sensing (proprioception and vision based control), and on a bimanual teleoperated robot for agricultural harvesting. One of our work has been accepted to IROS 2025.

Academic Projects (selected)

Nonlinear MPC for Flexible drones

Robotic Perception Group (RPG - UZH)

- This project aims to identify, integrate and then **develop a model of a flexible quadrotor** that strikes a balance between accuracy and efficiency. The approach combines model predictive control with **precise numerical integration** over a short initial horizon and a simplified, lower-fidelity model for longer-term planning.

Sep. 2025 - Jan. 2026

Multi Camera Gaussian Splat SLAM

Computer Vision and Geometry Group

- Me and my team synthesized a multi-camera SLAM system that uses **3D Gaussian Splatting** for robust localization, dense reconstruction, and novel view synthesis. By integrating **multiple fisheye cameras**, the system is designed to operate in challenging real-world environments and produce photorealistic maps. **Custom CUDA kernels** have been produced for the underlying NeRF NN.

Feb. 2025 - Jun. 2025

Monocular Visual Odometry

ETH Zurich

- Me and my team successfully implemented a **monocular visual odometry** pipeline as part of a course project. The system initializes 3D landmarks, tracks keypoints across frames, estimates camera poses using 2D-3D correspondences, and triangulates new landmarks to maintain localization accuracy.

Sept. 2024 - Jan. 2025

Academic Publications

- Petitti, A.**, Obayashi, N., Bianchi, G. & Hughes, J. *Camera-tracked Soft Underwater Robot Enabling Robust Orientation Control for Maneuverability*, **Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)**, Hangzhou, China, Oct. 2025.

Honors & Awards

Digital Twin Bootcamp Winner, My team's project was selected as the best one, among other 5 teams. I

Sept 2022 participated in the 2023 edition as tutor.

Cambridge, UK

NASA Space App Challenge: Global Nominee, Our project was selected from over 5500 teams from around the world to be part of the global challenge (top 10%).

Oct 2023

Rome, Italy

FKI - Federazione Karate Italia

Rome, Italy

COMPETITIVE KARATE ATHLETE, SELECTED RESULTS:

Nov 2018- Present

- 8th World Fudokan Karate Championship: Placed 2nd among 27 athletes from across the globe, who had been selected, by the relevant national teams to take part in this competition in the all belts, all styles category
- Italian absolute Championship Libertas (2023 and 2024): Placed 1st among 10 selected athletes from all over Italy to compete in the all belt all style category

Lead the Future mentorship

Rome, Italy

MENTEE

Sep. 2022 - Present

- Leading mentorship non-profit organization for top students in STEM, with acceptance rate below 20%, selected between 5000 applicants
- LTF empowers top-performing students to achieve their goals by giving them one-on-one guidance from high-impact mentors coming from the world's leading STEM innovation hubs such as Silicon Valley and CERN.